

Samad Amini | PhD | Applied AI Scientist

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EDUCATION

Boston University

September 2019 – September 2024

PhD candidate in Systems Engineering, GPA: 4.0/4.0 – Advisor: Ioannis Paschalidis.

Boston, MA

Sharif University of Technology

Sep 2015 – Sep 2018

Master of Science in Electrical Engineering, Control Systems.

Tehran, Iran

WORK EXPERIENCE

Machine Learning Intern | Philips Research, Cambridge, MA

Summer 2022

- **Designed a cloud-based pipeline** for precise landmark detection in 3D medical imaging, utilizing AWS and MONAI.
- **Enhanced runtime performance** through parallel processing and optimized data processing on cloud platforms.

Data Science Intern – NLP | Massachusetts General Hospital, Cambridge, MA

Spring 2022

- Developed a precise approach for cognitive assessment from clinical notes, **employing transformer-based models**, achieving 98% AUC.

TECHNICAL SKILLS

Programming: Python, C, Matlab, Perl, R, SQL.

Machine Learning Frameworks: PyTorch, TensorFlow, Scikit-Learn.

Computer Vision: OpenCV, MONAI, ResNet, U-Net.

Natural Language Processing: LLMs (Llama 3, LangChain), HuggingFace, SpaCy, BERT, NER.

Data Analysis: Pandas, NumPy, SciPy.

Cloud Computing: AWS, Google Cloud Platform.

RELEVANT PROJECT EXPERIENCE

LLM Pipeline for Stroke Severity Prediction from Clinical Notes

Sep 2023 – present

- Developed an LLM pipeline integrating **RAG, Langchain, and Chroma vectorDB** with Llama 2 for extracting critical information from stroke patient notes.
- Currently aiming at prediction of stroke severity from neurological exams and enhancing patient care management.

Statistical Analysis and Development of Predictive Models

Jan 2023 – Sep 2023

- **Predicted Alzheimer's Disease progression** in future within 6 years from speech using Transformer-based language models, achieving an accuracy of 78.2% and a sensitivity of 85.6%.
- Employed machine learning algorithms like **Random Forest** and **XGBoost** for Alzheimer's resilience identification.

Automating Cognitive Impairment Diagnosis from Speech/Text

Jan 2021 – Dec 2021

- **Developed an AI automated pipeline for dementia detection** with 96% AUC, enhancing predictive analysis in clinical research.
- Optimized feature selection in clinical research through recursive feature elimination techniques – **leading a team of five researchers**.

Predicting Dementia from Drawing

Jun 2020 – Dec 2020

- Pioneered a **Deep Learning project for dementia detection**, achieving a 92% AUC.
- **Led the project to the deployment** phase, successfully launched the model on <https://health-ai.bu.edu>.

SELECTED PUBLICATIONS (🌐 GOOGLE SCHOLAR)

- [S. Amini](#) et al., Prediction of Alzheimer's Disease Progression within 6 Years Using Speech: A Novel Approach Leveraging Language Models, Alzheimer's & Dementia, 2024.
- [S. Amini](#) et al., Automated Detection of Mild Cognitive Impairment and Dementia from Voice Recordings: A Natural Language Processing Approach, Alzheimer's & Dementia, 2022.
- [S. Amini](#) et al., An Artificial Intelligence-Assisted Method for Dementia Detection Using Images from the Clock Drawing Test, Journal of Alzheimer's Disease, 1-9, 2021.

GRADUATE LEVEL COURSES

Introduction to Learning from Data – Reinforcement Learning – Deep Learning – Optimization.